

WEI KUO

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Education

New York University, Abu Dhabi

Sep. 2021 – May 2025

Bachelor of Science in Computer Science & Minor in Mathematics

Abu Dhabi, UAE

Cumulative GPA: 3.94/4.00

Honors & Awards: Magna Cum Laude, NYU Abu Dhabi Founder's Award (Highest Honors)

Publications: Neutralizing Narratives: Debias News Articles with AI (First Author, Under Review at COLM'25)

Technical Skills

Core Languages: Python, C/C++, JavaScript, TypeScript

ML/AI Stack: PyTorch, TensorFlow, LangChain, Agentic AI, RAG, Claude Code

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Git, n8n, RPA

Data Tools: Pandas, NumPy, Excel, PowerBI, SQL, MongoDB, VBA

Work Experience

PicCollage

June 2025 – Now

AI & Automation Engineer

Taipei City, Taiwan

- Architected n8n automation workflows integrating 6 platforms, including Amplitude and AppFigures, with Excel and VBA scripting, eliminating manual data entry and saving 50 minutes per iteration
- Developed AI agents for dynamic bid optimization on Apple Search Ads, automatically adjusting keyword bids based on real-time CPA/ROAS metrics and improving ad spend efficiency across 200+ campaigns.
- Redesigned PicCollage's HRIS system with 4 interconnected databases to securely manage 300+ employees' data, automating cross-platform synchronization and reducing HR administrative overhead by 70%

Taiwan Semiconductor Manufacturing Company (TSMC)

June 2024 – Sep 2024

Machine Learning Intern

Tainan, Taiwan

- Implemented an XGBoost model that enhanced TSMC's production forecast accuracy by 4%, replaced TSMC's traditional prediction model and directly impacting over 10,000 products and 150+ customers.
- Developed a robotic process automation (RPA) pipeline with SQL, Excel, and PowerBI to automate the data extraction, transformation, and loading (ETL) process, reducing data processing time by 30%.

GitRoll

Jan 2024 – Aug 2024

Software Engineer Intern

Hong Kong, China

- Utilized the GitHub API for repository mining and extracted insights on essential libraries employed across diverse programming languages from 100+ repositories.
- Evaluated 200+ GitHub repositories with OpenAI's API to provide in-depth analysis.

Metamory Technology Limited

May 2023 – Jan 2024

Web Development Intern

Hong Kong, China

- Optimized react.js-based website for peak performance in GA4, GTM, and Google Ads.
- Developed 13 GTM tags, seamlessly integrated with GA4 for comprehensive user activity insights. Strategically targeted prospective audience on Google Ads, resulting in a remarkable 160% surge in web traffic.

Research

NYU Tandon's Flexible AI-enabled Mechatronic Systems Lab (FAMS)

June 2024 – May 2025

Head Researcher & Product Manager

New York Hybrid, USA

- Led a team of 9 to develop an AI-driven educational platform, connecting students and professors to 8 specialized experts across academic and wellness domains.
- Implemented automatic expert selection with semantic routing while ensuring security with NYU SSO & Kubernetes.
- Created vector databases from 1,000+ pages of course materials and mental health resource across 100+ documents.
- Mitigated hallucinations using Retrieval-Augmented Generation (RAG) techniques and vector embeddings.

NYU Abu Dhabi's Data Science and AI Lab

Feb 2024 – May 2025

Researcher & First Author

Abu Dhabi, United Arab Emirates

- Studied the racial and gender bias trends in 10 popular news publishers with over 355k articles from the past decade.
- Leveraged 6 state-of-the-art LLMs for bias detection and removal, employing few-shot learning and ensemble methods.
- Designed a large-scale crowdsourced study using randomized controlled trials to assess the impact of debiased news on readers' perceptions, applying causal inference techniques for analysis.

Certificates & Contests

TSMC Regional Intern Competition Gold Award (4/250 Interns) TSMC	August 2024
IBM Data Science Professional Certificate IBM	January 2024
Deep Learning Professional Certificate DeepLearning.AI	July 2023
American Region Mathematics League International 3rd	June 2020
Taiwan Region Mathematics League National Bronze	August 2020

Relevant Coursework

- Advanced Machine Learning
- Applied Machine Learning
- Software Engineering
- Probability & Statistics
- Database Design & Implementation
- Algorithmic Problem Solving

Projects

LinkedIn Weak Ties AI Assistant | *Python, FastAPI, React, ChromaDB, Azure OpenAI, Tailwind CSS* May 2025

- Built a two-tier semantic search system that identifies top 4 relevant connections from 1000+ LinkedIn profiles in under 100ms using multi-attribute vector search and GPT-4.1 refinement.
- Implemented incremental profile enrichment pipeline with async processing, handling 10 concurrent API requests with real-time progress tracking to process only new connections.
- Developed full-stack application with ChromaDB vector store using weighted similarity scoring across 4 profile dimensions (summary, position, location, industry) for intelligent connection matching.
- Integrated AI-powered outreach generator that creates personalized LinkedIn reconnection messages based on enriched profile context and user business goals.

Scalable AI Multi-Expert System | *Python, Langchain, ChromaDB, FastAPI, React, LLMs* April 2025

- Developed a multi-layer routing framework that automatically directs user queries to domain experts in $O(\log n)$.
- Deployed 11 cross-domain experts with 95% routing accuracy under 50ms routing time, processing knowledge from 184 documents containing over 5,000+ pages.
- Built a full-stack application that integrated vector database, RAG techniques for knowledge retrieval, and session management for conversation continuity.
- Created incremental learning pipeline that tracks file changes and automatically updates routing network.

AI-Powered Course Assistant | *Python, Langchain, ChromaDB, FastAPI, PyDantic, LLMs* Dec 2024

- Led backend development with FastAPI, Retrieval-Augmented Generation (RAG) techniques, and ChromaDB vector database to efficiently manage over 10,000 messages and 94 course documents and ensure fast and accurate responses.
- Utilized advanced scraping techniques to extract and process course materials, converting them into vector embeddings with OpenAI's embedding models and Implemented sentiment analysis for proactive moderation.
- Deployed the service on Kubernetes within NYU HSRN, implementing SSO authentication to secure access, seamless integration of microservices, and robust performance under high request volumes.

Online Telehealth Portal - Smartel | *Amazon AWS, FastAPI framework, React, PyDantic, Typescript* Jan 2024

- Smartel allows users to schedule appointments, record, transcribe, and generate consultation summaries, and provide physicians with concise and organized session reports.
- Led the backend development, implementing robust FastAPI and integrating OpenAI technologies (Whisper and GPT-4) to enable seamless conversation transcription and automated consultation summary generation.
- Designed a scalable database with Amazon AWS and backend architecture with PyDantic, FastAPI, SQLAlchemy, and PostgreSQL. Capable of handling 1000+ concurrent users, 500 bookings per minute, and response time of under 3 seconds for data retrieval operations.

Hotel Demand EDA & Prediction | *Python, C++, Pytorch, Pandas, Numpy, Plotly, Jupyter Notebook* Dec 2023

- Analyze historical hotel booking data with advanced predictive analytics techniques to forecast future booking patterns, providing actionable insights to drive strategic business decisions and optimize resource allocation.
- In-depth exploratory data analysis in customer demographics, pricing dynamics, market segmentation trends, etc.
- Spearheaded the development and evaluation of machine learning models including logistic regression, decision tree, k-nearest neighbors, and neural networks, achieving up to 85.10% accuracy in predicting hotel booking cancellations.

News Analysis Platform | *Flask, Python, SQL, Pandas, Matplotlib, Beautiful Soup* Nov 2023

- A full-stack web application using Flask framework and MySQL database to fetch, analyze, and visualize news data.
- Integrating NewsAPI and OpenAI's GPT Model to provide users with real-time news analysis and sentiment prediction.
- Developed a MySQL database for storing and managing large volumes of news articles and analysis results.